COSC 1030 SU25 Programming Assignment 5B: Distributed Data Computing

In this class we have previously introduced the topic of **parallel and distributed computing**. In this assignment we will explore the subtopic of **distributed data**. Often, computing applications use data that is located on a database that is not on the computer or even the network where the program is running. In this case, the application may need to interact with a remote service to obtain the data needed. These operations require learning to use libraries of functions, which is this assignment's other focus.

NOTE: This assignment assumes knowledge of cURL, JSON, and C++ functions. If you have not done the cURL and JSON assignments as well as the assigned reading on C++ functions, you should consider doing these activities first.

PART 1: Using the curl.h library in a C++ program (video for this section)

Unlike most of the assignments up to this point in the course, in this assignment you will be provided with starter code that will need to be altered to meet the specific requirements of this assignment. To get the first piece of starter code, from your Linux terminal window:

```
git clone https://github.com/Casper-College-COSC/NumbersAPI
```

This command will use git (a version control system) to clone (make a copy of) a repository that is residing on github at the address in the command.

The README folder has the information on compiling this code. Compile and run the code to see what it does. Note the addition of the -lcurl part of the compile command. This is because we are going to use a library that uses the cURL tool to retrieve some data from a repository via a web server.

This code was mostly written by chatGPT when I prompted it for code that would demonstrate how to use the NumbersAPI. An API is the set of rules and protocols for interacting with another software application. One of the pieces of data that can be requested by this particular service is a piece of trivia about a number. Look at the code and see if you can understand how this is being done.

TURN IN: nothing to to turn in for this section, it is instructional.

Part 2: Using the json.hpp library to interact with JSON formatted data in C++ (video for this section)

For this part of the assignment, get the code at the following repository:

```
git clone https://github.com/Casper-College-COSC/numbersAPIwJSON
```

This code also has a README with instructions for compiling. Compile and run the code to see what it does. Then take a look at the main.cpp file to see how the "json.hpp" library has been included. This is a library that was written primarily by Niels Lohmann. If you visit the json library repository, you can find out that it is an open source project that has been contributed to by over 300 other programmers.

TURN IN: nothing to turn in for this section, it is instructional

Part 3: Develop a program that uses the USGS API to obtain information about seismic activity and process the records to report activity of interest.

The United States Geological Survey (USGS) records seismic activity using a network of seismic stations. The data is processed and made available for use by anyone. One of the ways that access to this database is provided is through a JSON object as described here:

https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php

Take a look at some of the JSON records on the links on the right side of this page. Which one of these links would be appropriate for reporting earthquakes above a certain magnitude for the previous month?

Modify the code given in parts 1 and 2 to practice retrieving and accessing different values from the database.

Specifications: The finished program output should be similar to the sample given below:

```
charlottegruner@Charlottes-MacBook-Pro Lab8c % ./geo_data
Enter the magnitude: 6.0

Finding earthquakes greater than 6 magnitude.

"Scotia Sea" 6.6****** TSUNAMI!! ******
"71 km ESE of Sarangani, Philippines" 6.1
"southern Mid-Atlantic Ridge" 6.2
"east of the Philippine Islands" 6.2
"83 km ESE of Nemuro, Japan" 6
"Kuril Islands" 6
"15 km NE of Paratebueno, Colombia" 6.3
"western Indian-Antarctic Ridge" 6.2
"46 km SW of Diego de Almagro, Chile" 6.4****** TSUNAMI!! ******

9 earthquakes were found. The largest magnitude was 6.6 located at "Scotia Sea"
```

HINT:

You will find that you will need a way to iterate through the “features” array in the json object to examine each of the “mag” values. An easy way to do is to take a look at each element in the array is to use a range based for loop combined with auto for automatically detecting the data type for each item in the array.

```
json response = json::parse(readBuffer);
for (auto element: response["features"]) {
   /* code to examine each element in “features” array */
}
```

Recall that `auto` will automatically choose the data type for `element`. And that the range-based for loop will iterate through each item of the “features” array.

TURN IN:

1. your .cpp file
2. a 2-3 minute video showing, explaining, and running the code.
3. A one paragraph reflection on this assignment. Was it too hard? Too easy? Interesting? Challenging? What would you change about this assignment to make it better?